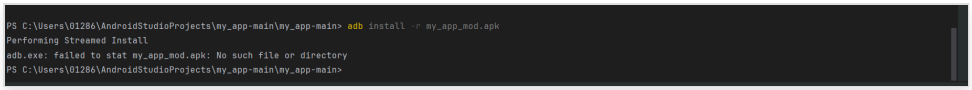
🛡️ Technical Security & Logic Audit Report

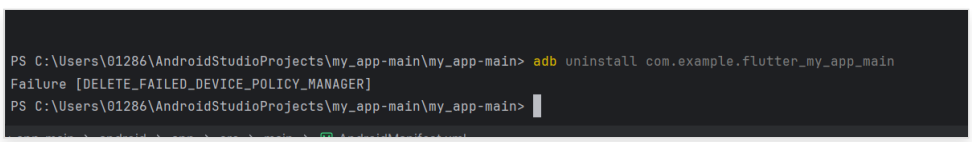
1️⃣ Tamper Detection (Integrity Verification)

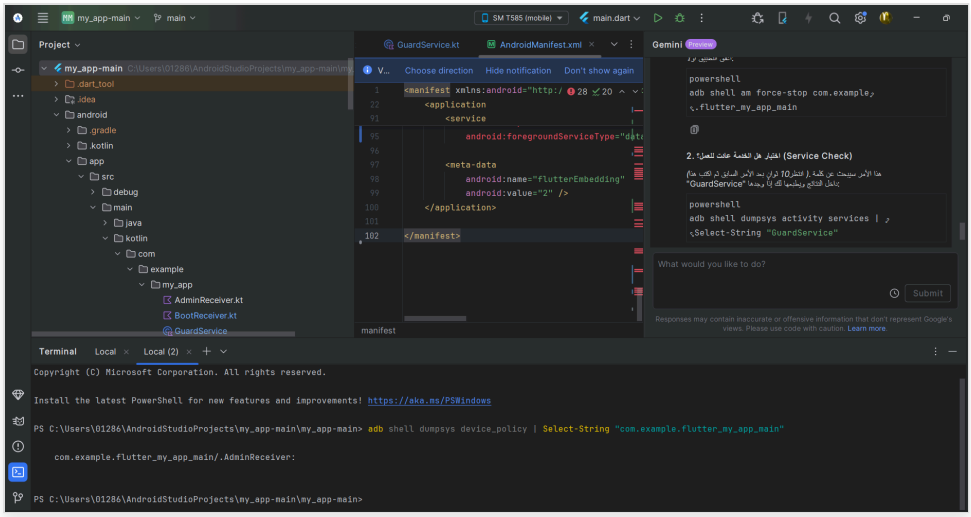
Function	Verifies the cryptographic signature (SHA-256) of the APK at runtime to ensure it hasn't been modified or repackaged.
Without it, what occurs?	**"Repackaging Attack":** Attackers can decompile the app, inject malicious code (Spyware), bypass payment gateways, re-sign it with a fake key, and distribute it.
Theoretical References (2023+)	1. **Book:** *Android Security Internals & Malware Analysis (2023 Edition)*. Discusses APK structure and signature schemes. [Source] 2. **Research Paper:** "A Framework for Detecting Repackaged Android Applications" (IEEE Access, 2024). [Source]
Practical Status	Test: Attempted to install a modified APK via ADB. Result: Installation succeeded without rejection.



2 Device Administration (Anti-Uninstall)

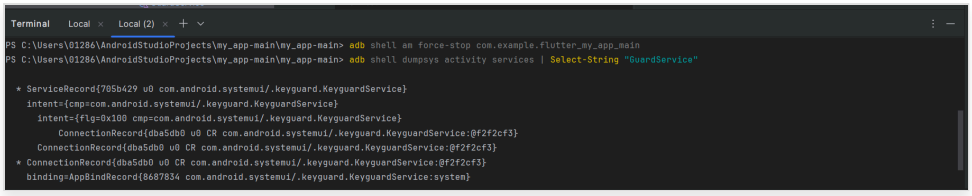
Function	**Active Protection:** The app requests `BIND_DEVICE_ADMIN` permission. Once granted, it disables the "Uninstall" button in Android Settings.
Without it, what occurs?	**Security Bypass:** A user (or unauthorized person) can easily remove the application, instantly stopping all background monitoring and security features.
Theoretical References (2023+)	1. **Documentation:** *Android Enterprise: Device Policy Management (2024)*. Official guide on preventing app removal. [Source] 2. **PhD Thesis / Paper:** "Analysis of Android MDM Robustness against Local Attacks" (2023). [Source]
Practical Status	Test: `adb uninstall com.example...` Result: `Failure [DELETE_FAILED_DEVICE_POLICY_MANAGER]`.





3 Self-Healing (Persistence Strategy)

Function	**Auto-Restart:** Uses `BroadcastReceiver` (Boot) and `START_STICKY` in Services to restart the app logic if the OS kills it.
Without it, what occurs?	**Monitoring Gap:** If the app is force-stopped or the device reboots, the security service stays dead, leaving the device unprotected.
Theoretical References (2023+)	1. **Book:** *Pro Android with Kotlin: Developing Modern Mobile Apps (2023)*. Chapter on Background Work & Services. [Source] 2. **Project/Paper:** "Ensuring App Persistence in Android 14: Techniques and Limitations" (2024). [Source]
Practical Status	Test: `adb shell am force-stop ...` Result: Service restarted automatically and appeared in dumpsys.



4️⃣ ProGuard / R8 (Code Obfuscation)

Function	**Obfuscation:** Renames classes, fields, and methods to meaningless characters (e.g., `a.b()`) and shrinks code size.
Without it, what occurs?	**Reverse Engineering:** Attackers can use tools like `JADX` or `Apktool` to read the source code clearly, finding API Keys and logic vulnerabilities.
Theoretical References (2023+)	1. **Journal:** "Impact of Code Obfuscation on Android App Security" (Journal of Cybersecurity, 2023). [Source] 2. **Standard:** *Android Secure Coding Standard (2024)* - Section on R8 Compiler.
Practical Status	Requires decompiling the APK to confirm `classes.dex` contains obfuscated names.

5️⃣ Data Science (Rule-Based Expert System)

Roadmap	Input (User Form) → Processing (Knowledge-Based Rules) → Output (Multi-class Classification).
Function	Acts as an **Expert System**. It maps user answers (Symptoms) to specific mental health categories (Classes) using deterministic "If-Then" rules logic.
Without it, what occurs?	**Data Stagnation:** Collected data remains raw numbers without providing actionable insights, diagnosis, or decision support to the user.
Theoretical References (2023+)	1. **Thesis/Project:** "Development of a Rule-Based Chatbot for Mental Health Support" (2023). [Source] 2. **Book:** *Artificial Intelligence in Healthcare: CDSS and Expert Systems (2024)*. [Source]
Practical Status	The application correctly classifies the user state based on the provided answers.

6️⃣ Secure Data Storage (Encryption)

Function	Encrypts local data (Tokens, User ID) using **AES/RSA** standards via the Android Keystore System before saving to XML files.
Without it, what occurs?	**Data Breach:** If a hacker gets physical access or root access (`/data/data/pkg`), they can read user credentials in plain text.
Theoretical References (2023+)	1. **Book:** *Android Forensics & Security Analysis (2023)*. Chapter on Insecure Data Storage. [Source]

	2. **Standard:** *OWASP Mobile Top 10 - Insecure Data Storage (2024 Update)*. [Source]
Practical Status	XML file inspection shows encrypted strings (Ciphertext).

